ICT2403 – Graphics and Image Processing

Submission 03 – Frame Enhancement Method

1. Introduction to Frame Enhancement

Image enhancement is an important step in computer vision systems that improves the visual quality of images before performing further analysis such as segmentation or object detection. In this project, the goal is to detect road surface damages such as cracks and potholes from recorded road videos. However, raw video frames often contain various degradations that make damage detection difficult.

Frame enhancement is required to improve image visibility, remove noise, and increase contrast so that the target regions of interest can be identified more accurately.

In road damage detection systems, enhancement plays a significant role because road defects usually appear as small intensity variations on the road surface. If the image quality is poor, these defects may not be visible or may be incorrectly detected.

Raw video frames recorded using mobile cameras typically contain several challenges such as:

- Sensor noise generated by the camera sensor
- Compression noise caused by MP4 video encoding
- Uneven illumination due to sunlight variations
- Shadows from trees, vehicles, or buildings
- Motion blur caused by vehicle movement or handheld recording

These degradations can negatively affect the segmentation stage by hiding important structural details of the road surface.

Therefore, an image enhancement pipeline was developed in this project to improve frame quality through several stages including noise removal, contrast

enhancement, and sharpening. The enhanced frames provide clearer visual information, which improves the accuracy of the subsequent road damage detection process.

2. Identification of Frame Degradations

Before applying enhancement techniques, it is important to analyze the degradations present in the recorded video frames.

The recorded road video was examined visually and through histogram analysis to identify common image quality problems.

2.1 Noise Analysis

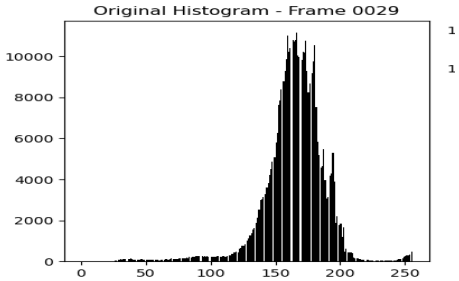
Noise is one of the most common degradations observed in digital images captured using mobile cameras. Sensor noise appears as random intensity variations across pixels, especially in low-light conditions.

In this project, two types of noise were observed:

- **Sensor noise** generated by the mobile camera sensor
- **Compression noise** introduced during MP4 video compression

These noise patterns appear as small pixel fluctuations that reduce image clarity and hide fine road crack structures.

To analyze noise, grayscale frames were extracted and histograms of pixel intensities were plotted. Noisy images typically show irregular histogram distributions due to random pixel variations.



Noise significantly affects the identification of road surface cracks because it can be mistaken as structural features during segmentation.

To reduce this problem, **Gaussian filtering and Median filtering** were applied during the enhancement stage.

2.2 Illumination Variations

Illumination variations occur due to changing lighting conditions during video recording. In outdoor road scenes, lighting conditions can change rapidly due to sunlight, shadows, or environmental factors.

The following illumination problems were observed:

- Uneven brightness across the road surface
- Shadows from trees and vehicles
- Overexposed bright regions caused by sunlight
- Underexposed dark areas

These illumination variations reduce contrast and make cracks less visible.

Histogram analysis of these frames showed uneven distribution of intensity values.

To address this issue, contrast enhancement techniques such as **contrast stretching and CLAHE (Contrast Limited Adaptive Histogram Equalization)** were applied.

2.3 Motion Blur and Camera Instability

Since the video was recorded using a moving vehicle or handheld camera, some frames exhibit motion blur. Motion blur reduces image sharpness and weakens edge structures that represent cracks or potholes.

Blurred images contain smoothed edges and reduced high-frequency components.

To analyze sharpness degradation, Laplacian-based sharpness measurements were used.

To improve image clarity, **image sharpening techniques** were applied using convolution kernels to enhance edge structures.

3. Pre-processing of Video Frames

Before applying enhancement techniques, the recorded road video was converted into individual image frames that can be processed using image processing algorithms.

3.1 Frame Extraction Method

The recorded road video was first converted into grayscale format using OpenCV. Grayscale conversion simplifies the image by reducing the three color channels (RGB) into a single intensity channel. This reduces computational complexity while preserving structural information such as cracks and potholes.

The grayscale conversion was performed using the OpenCV function:

```
cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
```

3.2 Frame Extraction Strategy

After converting the video to grayscale, individual frames were extracted from the video stream.

To reduce redundancy and computational load, **every 10th frame** was selected for analysis. This sampling strategy ensures that the dataset contains diverse frames while avoiding excessive processing of similar frames.

Extracted frames were saved in **PNG format** using the following naming structure:

frame_0.png
frame_1.png
frame_2.png

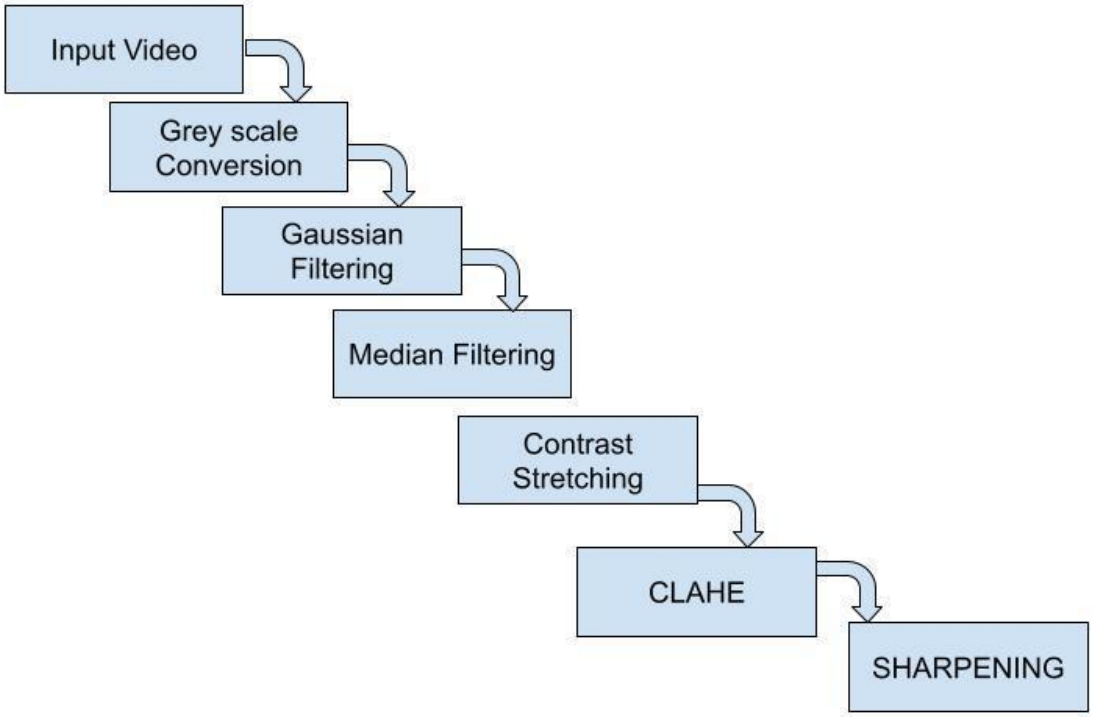
3.3 Frame Format and Resolution

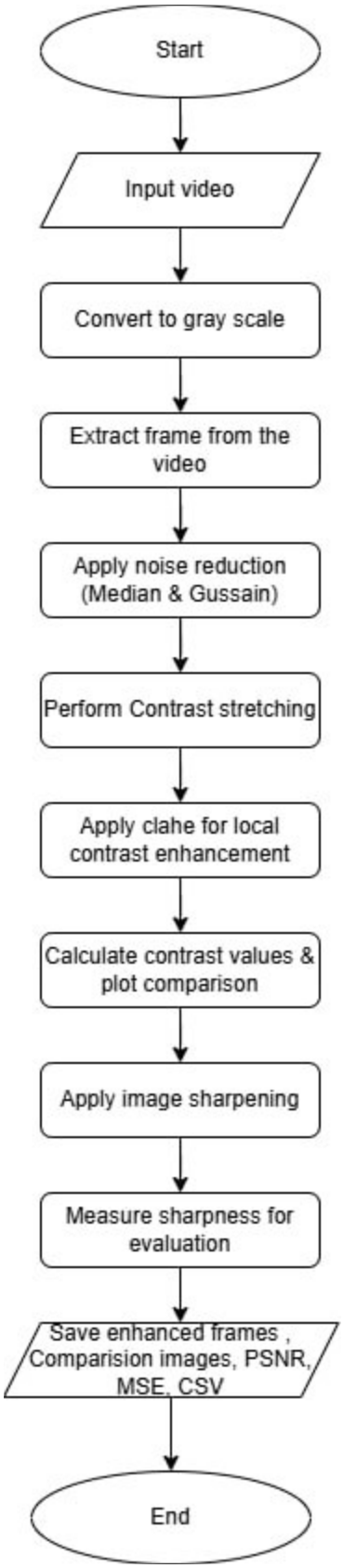
All extracted frames were stored in **PNG format** because it is a lossless image format that preserves image quality without compression artifacts.

The resolution of the frames remained the same as the original video resolution.

4. Proposed Image Enhancement Pipeline

The proposed enhancement pipeline progressively improves frame quality by reducing noise, enhancing contrast, and preserving structural edges, thereby providing suitable input frames for the subsequent road damage segmentation stage




Start
Input video
Convert to gray scale
Extract frame from the video
Apply noise reduction (Median & Gussain)
Perform Contrast stretching
Apply clahe for local contrast enhancement
Calculate contrast values & plot comparison
Apply image sharpening
Measure sharpness for evaluation
Save enhanced frames , Comparision images, PSNR, MSE, CSV
End

Input frame(color)

Grayscale frame

Denoised frame

Contrast stretched frame

CLAHED frame

Sharped frame

4.1 Noise Removal

Noise removal was performed using two filtering techniques:

Gaussian Filter

Gaussian filtering smooths the image by averaging neighboring pixels based on a Gaussian distribution.

Mathematically, the Gaussian filter is represented as:

$G(x,y) = (1 / 2\pi\sigma^2)\ e^{(-(x^2+y^2)/2\sigma^2)}$

A **5×5 Gaussian kernel** was used in the implementation.

```
cv2.GaussianBlur(image,(5,5),0)
```

Median Filter

Median filtering replaces each pixel value with the median value of neighboring pixels. This method is highly effective for removing **salt-and-pepper noise** while preserving edges.

```
cv2.medianBlur(image,5)
```

4.2 Contrast Enhancement

After noise removal, contrast enhancement was applied to improve visibility of road features.

Contrast Stretching

Contrast stretching expands the intensity range of the image to improve visibility.

The transformation is defined as:

Output = (Input − Min) / (Max − Min) × 255

This operation increases the difference between dark and bright regions.

CLAHE Enhancement

CLAHE (Contrast Limited Adaptive Histogram Equalization) enhances local contrast by dividing the image into small tiles and applying histogram equalization to each tile.

This method prevents over-amplification of noise and produces more natural contrast.

Implementation used:

cv2.createCLAHE(clipLimit=2.0, tileGridSize = (16,16))

4.3 Image Sharpening

Sharpening enhances edge structures using a convolution kernel:

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

This kernel enhances high-frequency components and improves crack visibility.

GitHub Repository

The complete implementation of the frame enhancement pipeline, including preprocessing, denoising, contrast enhancement, and sharpening algorithms, has been uploaded to the GitHub repository for collaborative development.

GitHub Link:

https://github.com/Ajanthan04/IP_Project

5.1 Test Data Preparation

The dataset used for testing was generated from recorded road videos.

Frames were extracted using the frame extraction method described earlier. A total of **30 frames** were selected for experimental evaluation.

Each frame underwent the following preprocessing steps:

- Grayscale conversion
- Frame extraction
- Noise removal
- Contrast enhancement
- Sharpening

5.2 Evaluation Metrics

Two quantitative metrics were used to evaluate the enhancement performance:

Mean Squared Error (MSE)

MSE measures the average squared difference between original and processed images.

$MSE = (1/N) \Sigma (I - K)^2$

Lower MSE values indicate better image quality.

Peak Signal-to-Noise Ratio (PSNR)

PSNR measures the ratio between maximum signal power and noise.

$PSNR = 20 \log10 (255 / \sqrt{MSE})$

Higher PSNR values indicate better image quality.

5.3 Testing the Enhancement Pipeline

The pipeline was tested using 30 extracted frames.

The following parameters were used:

Gaussian filter size = 5×5
Median filter kernel size = 5
CLAHE clip limit = 2.0
Tile grid size = 16×16

5.4 Presentation of Results

The enhancement results are presented using:

- Visual comparison images
- Histogram plots
- Quantitative evaluation tables

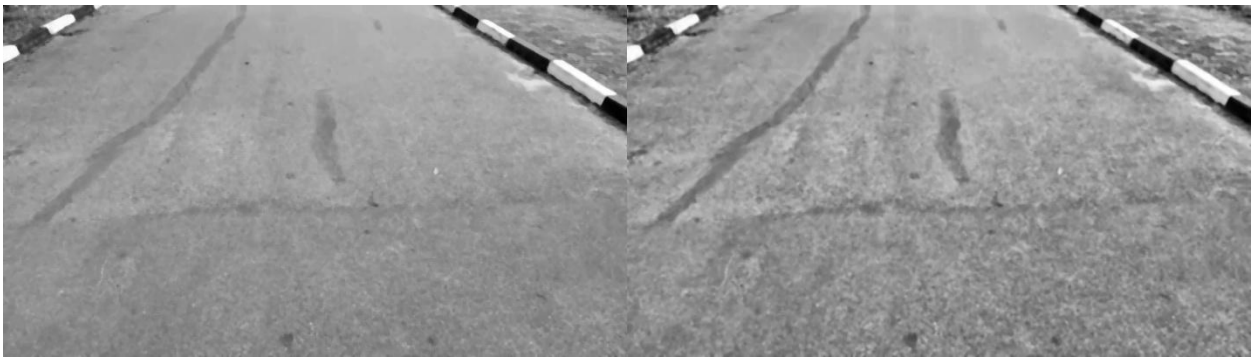
• Visual comparison images

Noise removal comparison: Original vs Gaussian vs Median

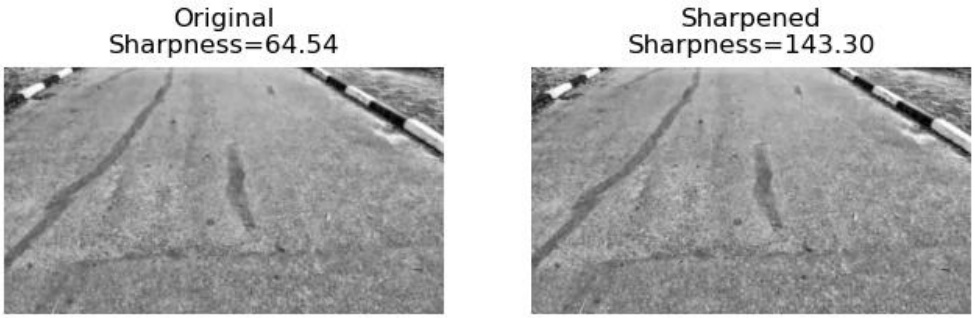
Contrast enhancement: Contrast stretching only

Adaptive enhancement: CLAHE

CLAHE side-by-side comparison with contrast stretched frame

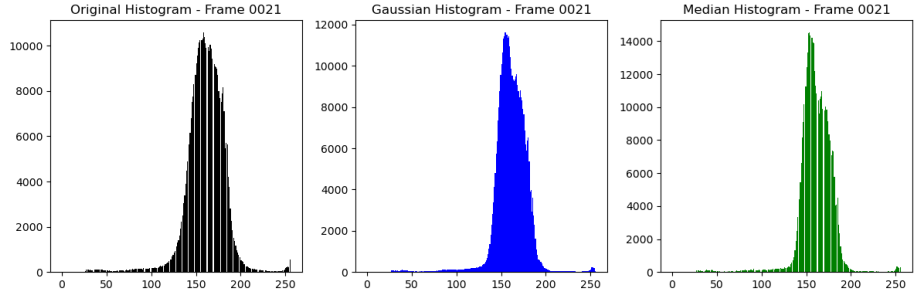


Sharpening: Original vs Sharpened frame

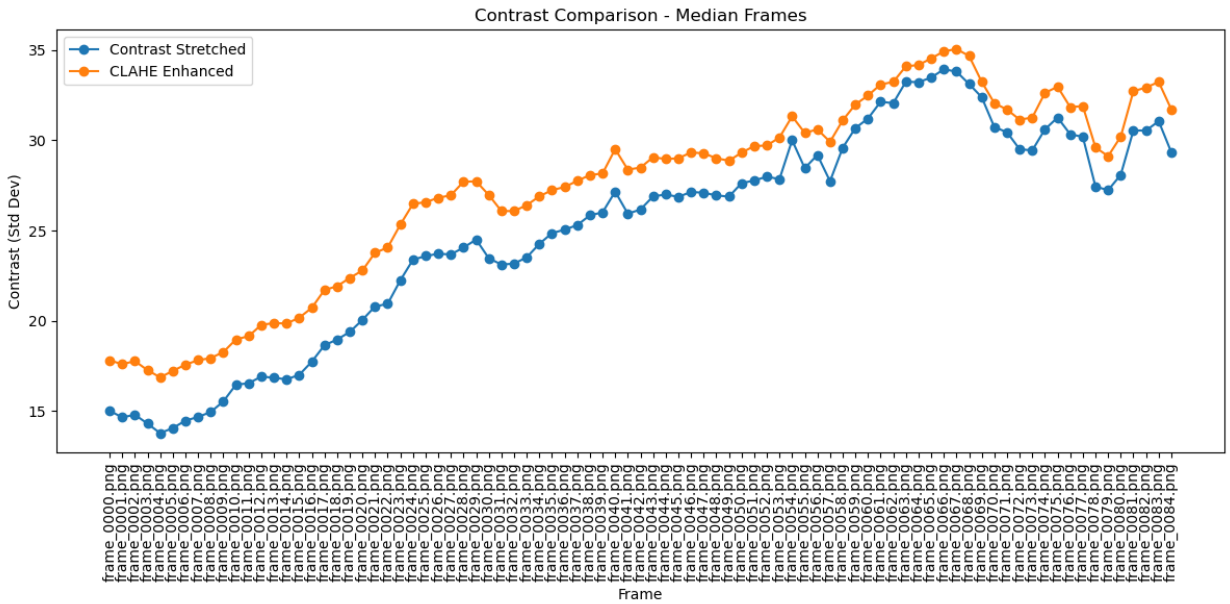


• **Histogram plots**

Histogram comparison of Original, Gaussian, Median



Contrast comparison plot (Contrast stretched vs CLAHE)



This demonstrates quantitatively that **contrast increased after enhancement.**

• <u>Quantitative evaluation tables</u>

PSNR & MSE for first 10–15 frames (Gaussian vs Median)

c	PSNR_Gaussian	PSNR_Median	MSE_Gaussian	MSE_Median
0	29.87	29.71	66.95	69.56
1	31.92	31.08	41.83	50.73
2	32.16	31.23	39.53	48.93
3	32.34	31.36	37.97	47.54
4	34.43	33.02	23.46	32.47
5	33.09	32.07	31.89	40.39
6	33.54	32.35	28.75	37.88
7	33.62	32.5	28.25	36.57
8	32.46	31.57	36.9	45.3
9	32.99	31.92	32.67	41.8
10	32.42	31.51	37.24	45.94
11	32.75	31.69	34.53	44.05
12	32.12	31.27	39.91	48.55
13	32.58	31.54	35.94	45.57

5.5 Interpretation of Results

The enhancement pipeline significantly improved frame quality. Noise reduction filters removed random pixel variations, while contrast enhancement increased visibility of road surface textures.

Sharpening further improved edge clarity, making cracks and potholes more distinguishable.

The combined enhancement techniques produced clearer frames suitable for segmentation and damage detection.

6. Recommendation

6.1 Final Recommendation

Based on experimental results, the enhanced frames provide improved contrast, reduced noise, and clearer edge structures. Therefore, these enhanced frames are suitable for the subsequent segmentation stage.

6.2 Effectiveness of Enhancement Techniques

Among the implemented techniques:

- Median filtering showed better performance for removing impulse noise.
- CLAHE significantly improved local contrast.
- Sharpening enhanced edge clarity.

6.3 Limitations

Some limitations were observed:

- Strong shadows may still affect visibility.
- Extreme lighting conditions may cause over-enhancement.
- Motion blur cannot be completely removed.

6.4 Impact on Segmentation Accuracy

Improved contrast and reduced noise help segmentation algorithms identify cracks more accurately. Edge enhancement also improves the detection of small structural details.

Future work may include advanced techniques such as deep learning-based enhancement.

7. Team Contribution Table

Member Name (Reg No: and Index No:	Degradation Analysis Responsibility (Enhancement Pipeline Stage)	Coding Contribution (Algorithm Done, Coding Segment)	Experimental Task (Test Preparation)	Result & Interpretation Contribution	Evidence to Attach (Qualitative or Quantitative or Both)
Member1 ICT/2023/118	frame extraction and preprocessing Noise analysis (sensor noise, compression noise) using histogram and visual inspection.	Implement noise removal (Median/Gaussian filtering)	Test noise reduction on 30 extracted frames	Analyze noise reduction effectiveness and impact on crack visibility	Before–after images, histograms, code commits
Member2 ICT/2023/117	Illumination variation analysis (shadows, uneven lighting, over/underexposure)	Implement contrast stretching and histogram equalization	Run enhancement on low-light and shadow frames	Interpret contrast improvement and ROI visibility	Comparison figures, contrast plots
Member3 ICT/2023/086	Motion blur and camera instability analysis (frame sharpness evaluation)	Implement sharpening and smoothing techniques	Test enhancement on blurred frames	Evaluate edge preservation and clarity improvement	Edge comparison images, parameter settings
Member4 ICT/2023/119	Overall degradation summary	Develop full enhancement pipeline integration and flowchart	Execute complete pipeline testing and metric calculation	Final result interpretation and recommendation for segmentation stage	Flowchart, tables of results, GitHub link

8.Individual Responsibilities

<u>Member 1 – Noise Analysis, Frame Preprocessing, and Extraction</u>

Tools Used: Python, OpenCV, Matplotlib, NumPy, CSV, Pandas

Member 1 focused on preparing the raw video frames for analysis and reducing noise to improve clarity for further processing.

Frame Extraction and Preprocessing:

- Converted the original video (Video02.mp4) to grayscale to simplify processing. Saved as road_video_grayscale.mp4.
- Extracted every 10th frame from the grayscale video to reduce processing load while keeping sufficient frames for analysis.
- Saved extracted frames as PNG images in the folder frames/.

Noise Removal:

- Applied Gaussian filter (cv2.GaussianBlur) to smooth frames.
- Applied Median filter (cv2.medianBlur) to reduce noise while preserving edges.
- Filtered frames saved in separate folders:
    - denoised_gaussian/ → Gaussian-filtered frames
    - denoised_median/ → Median-filtered frames

Visual Comparison Images:

- Side-by-side comparisons of original grayscale frames with Gaussian and Median-filtered frames for the first 30 frames.
- Saved in comparison_results/.
- Observation: Gaussian smooths noise but slightly blurs edges; Median preserves edges while reducing noise, making cracks more visible.

Histogram Plots:

- Pixel intensity histograms for original, Gaussian, and Median frames.
- Saved in histogram_results/.
- Observation: Median filtering produces smoother distributions while maintaining structural details.

Quantitative Evaluation:

- PSNR and MSE metrics calculated for 30 frames and saved in denoising_metrics.csv.
- Observation: Median filtering achieves higher PSNR and lower MSE, confirming better edge-preserving denoising.

Folders for Reference: frames/, denoised_gaussian/, denoised_median/, comparison_results/, histogram_results/, denoising_metrics.csv.

<u>Member 2 – Illumination Enhancement</u>

Tools Used: Python, OpenCV, NumPy, Pandas, Matplotlib

Member 2 focused on improving frame visibility under uneven lighting and shadows.

Visual Comparison Images:

- Comparison of contrast-stretched frames (enhanced_gaussian_contrast/, enhanced_median_contrast/) with CLAHE-enhanced frames (enhanced_gaussian_clahe/, enhanced_median_clahe/).
- Side-by-side images saved in comparison_images_gaussian/ and comparison_images_median/.
- Observation: CLAHE improves contrast in low-light and shadowed regions, making road damages more visible.

Histogram / Contrast Plots:

- Standard deviation of pixel intensities plotted for contrast-stretched and CLAHE-enhanced frames, saved in contrast_plots/.
- Observation: Enhanced frames show measurable contrast improvement, particularly in darker areas.

Quantitative Evaluation:

- Contrast values recorded in contrast_values_gaussian.csv and contrast_values_median.csv.
- Observation: CLAHE consistently increases contrast compared to the original and contrast-stretched frames.

Folders for Reference: enhanced_gaussian_contrast/, enhanced_median_contrast/, enhanced_gaussian_clahe/, enhanced_median_clahe/, comparison_images_gaussian/, comparison_images_median/, contrast_plots/, contrast_values_gaussian.csv, contrast_values_median.csv.

Member 3 – Motion Blur and Sharpening

Tools Used: Python, OpenCV, NumPy, Matplotlib, Pandas

Member 3 focused on improving sharpness and mitigating motion blur in the frames.

Visual Comparison Images:

- Original frames converted to grayscale and Gaussian-smoothed, then sharpened using a convolution kernel.
- Side-by-side comparisons of original vs sharpened frames saved in COMPARISON_RESULTS/.
- Observation: Sharpened frames show improved edge clarity, enhancing visibility of cracks and surface irregularities.

Quantitative Evaluation:

- Sharpness metrics computed using Laplacian variance, saved in sharpness_values.csv.
- Observation: Sharpened frames consistently exhibit higher sharpness values, confirming improved edge definition.

Folders for Reference: SHARP_MC/, SHARP_GC/, COMPARISON_RESULTS/, sharpness_values.csv.

<u>Member 4 – Full Pipeline Integration</u>

Tools Used: Python, OpenCV, Matplotlib, NumPy, Pandas

Member 4 focused on integrating the complete enhancement pipeline and preparing frames for final testing.

Visual Comparison Images:

- Coordinated outputs from all stages, showing final enhanced frames ready for segmentation.
- Observation: Frames after noise removal, contrast enhancement, and sharpening are significantly clearer and more suitable for crack and road damage detection.

Quantitative Evaluation:

- Summarized metrics from PSNR, MSE, contrast, and sharpness for all frames, providing a complete view of enhancement effectiveness.

Folders for Reference: All processed frames folders from Members 1–3 (denoised_gaussian/, denoised_median/, enhanced_gaussian_clahe/, enhanced_median_clahe/, SHARP_MC/, SHARP_GC/), comparison images (comparison_results/, comparison_images_gaussian/, comparison_images_median/, COMPARISON_RESULTS/), and metrics CSV files (denoising_metrics.csv, contrast_values_gaussian.csv, contrast_values_median.csv, sharpness_values.csv).

Submission03_GROUP 24

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